SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	VIVETHA B	**Reg. No.:**	192424170
Section / Batch:	B.TECH (AI&DS)	**Date:**	

Code of Conduct

*I **VIVETHA 192424170** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **VIVETHA B**

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus: *"Data science is powerful, data science drives innovation, data science is evolving."*
The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences. The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- λ_1 = 0.5 (Trigram)
- λ_2 = 0.3 (Bigram)
- λ_3 = 0.2 (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

1. MLE of Bigram Probability P(science | data)

The MLE formula for a bigram is:

P(science | data) = C(data science) / C(data)

Given:

- C(data science) = 3
- C(data) = 3

Therefore:

P(science | data) = 3 / 3 = 1.0

Answer: P(science | data) = 1.0 or 100%

Interpretation:
The probability is **1.0**, meaning that in the given corpus, every occurrence of **data** is followed by **science**. Therefore, when a user types **"data"**, the keyboard has very high confidence in predicting **"science"** as the next word.

2. Backoff Model for "data science improves"

The sequence **"data science improves"** is unseen because:

C(data science improves) = 0

Therefore, the trigram probability cannot be estimated directly.

The backoff model moves to lower-order n-grams:

Trigram → Bigram → Unigram

For **"improves"**:

- `science improves` is unseen.
- `improves` is also not present in the given corpus.

Therefore, using the available corpus:

P(improves | data science) ≈ 0

Hence:

P(data science improves) ≈ 0

Why backoff is essential:
In real-time keyboard prediction, users constantly type new or uncommon word combinations. A higher-order n-gram may not have seen the exact sequence. Backoff allows the system to use lower-order information instead of completely failing when an unseen sequence occurs.

3. Deleted Interpolation for "data science is"

The interpolation formula is:

$$\mathbf{P(w_3 \mid w_1w_2) = \lambda_1P_3(w_3 \mid w_1w_2) + \lambda_2P_2(w_3 \mid w_2) + \lambda_3P_1(w_3)}$$

Given:

- $\lambda_1 = 0.5$
- $\lambda_2 = 0.3$
- $\lambda_3 = 0.2$

For **"data science is"**:

The trigram probability is:

P(is | data science) = C(data science is) / C(data science)

From the corpus:

- C(data science) = 3
- C(data science is) = 2

Therefore:

P_3 = 2/3 = 0.667

The bigram probability is given:

$P_2 = P(\text{is} \mid \text{science}) = 0.66$

For the unigram probability, the corpus has 17 words in total and **is** occurs 2 times:

$P_1(\text{is}) = 2/17 \approx 0.118$

Therefore:

$P = (0.5 \times 0.667) + (0.3 \times 0.66) + (0.2 \times 0.118)$

$P = 0.3335 + 0.198 + 0.0236$

$P \approx 0.555$

Answer: P(data science is) ≈ 0.555

Interpretation:
Deleted interpolation combines **trigram, bigram, and unigram probabilities**. The trigram provides detailed context, the bigram provides additional coverage, and the unigram provides a fallback. This makes the model more reliable when training data is sparse.

4. Entropy of P(is) = 0.66 and P(drives) = 0.33

Entropy is calculated using:

$H = -\Sigma P(x) \log_2 P(x)$

Therefore:

$H = -[0.66 \log_2(0.66) + 0.33 \log_2(0.33)]$

Approximately:

$H \approx 0.66(0.599) + 0.33(1.599)$

$H \approx 0.395 + 0.528$

$H \approx 0.923$ **bits**

Answer: Entropy ≈ 0.92 bits

Interpretation:
The entropy of approximately **0.92 bits** indicates a relatively limited amount of uncertainty. The keyboard strongly prefers **"is" (66%)** over **"drives" (33%)**, so its prediction confidence is reasonably high.

In an intelligent keyboard:

- **Low entropy** → predictions are more certain.
- **High entropy** → predictions are more uncertain.
- Entropy helps evaluate how confidently the language model can predict the next word.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.
Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.
Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

1. POS Tagging

Sentence 1: "Book a flight ticket now."

Book/VB a/DT flight/NN ticket/NN now/RB

- **Book → VB** (Verb), because it is an instruction or command.
- **a → DT** (Determiner)
- **flight → NN** (Noun)
- **ticket → NN** (Noun)
- **now → RB** (Adverb)

Here, **"Book" is a Verb (VB)** because it means the action of reserving a flight.

Sentence 2: "This book is interesting."

This/DT book/NN is/VBZ interesting/JJ

- **This → DT** (Determiner)
- **book → NN** (Noun)
- **is → VBZ** (Verb)
- **interesting → JJ** (Adjective)

Here, **"book" is a Noun (NN)** because it refers to a book and is used as the subject of the sentence.

Therefore:

Book/VB → "Book a flight."

book/NN → "This book is interesting."

The POS tag changes according to the **context**.

2. HMM Probability for "Book" as Verb

Given:

P(book | VB) = 0.6

P(Start → VB) = 0.5

The HMM probability is:

P(VB, book) = P(Start → VB) × P(book | VB)

= 0.5 × 0.6

= 0.30

Therefore:

ANSWER: Probability of "book" as VB = 0.30

For comparison:

P(NN, book) = P(Start → NN) × P(book | NN)

= 0.5 × 0.4

= 0.20

Since:

0.30 > 0.20

the HMM favors **VB**.

This agrees with the context because **"Book a flight ticket now"** is a command, so **Book** functions as a verb.

3. Rule-Based Tagging vs Stochastic HMM Tagging

Rule-Based Tagging uses manually created linguistic rules to assign POS tags. For example, a rule may identify **"Book"** as a verb when it appears at the beginning of a command.

Stochastic or HMM Tagging uses probabilities learned from language data. It calculates which POS tag is more likely based on the word and its context.

For example:

Book a flight → VB

This book → NN

For a large-scale chatbot, **Stochastic/HMM tagging is more suitable** because it can handle large amounts of language data and resolve ambiguous words using probabilities.

A combination of **rule-based and statistical methods** can provide even better performance.

4. Role of POS Tagsets and Word Classes

Standardized POS tagsets such as the **Penn Treebank POS Tagset** provide consistent grammatical labels for words.

Examples:

NN → Noun

VB → Verb

JJ → Adjective

RB → Adverb

DT → Determiner

POS tags help the chatbot understand the grammatical structure of user messages.

For **intent detection**, POS information helps identify actions. For example:

"Book a flight"

The **VB** tag indicates an action, helping the chatbot identify the **flight booking intent**.

For **response generation**, POS information helps the system understand sentence structure and generate appropriate responses.

POS tagging also helps with **lexical ambiguity**. The word **"book"** can be a noun or a verb depending on its context.

Therefore, standardized POS tagsets improve:

- **Intent detection**
- **Context understanding**
- **Word ambiguity resolution**
- **Entity extraction**
- **Response generation**
- **Overall chatbot accuracy**

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
   - economic (120 occurrences)
   - growth (450 occurrences)
   - increases (210 occurrences)
   - employment (380 occurrences)

   Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

1. Apply the Transformation Rule

Given initial POS tags:

economic/JJ growth/NN increases/NNS employment/NN

The transformation rule is:

Change NNS to VBZ if the preceding word is tagged as NN.

The word **"increases"** is tagged as **NNS** initially.

Its preceding word is **"growth"**, which is tagged as **NN**.

Therefore, the rule applies:

increases/NNS → increases/VBZ

Corrected POS sequence:

economic/JJ growth/NN increases/VBZ employment/NN

The correction is linguistically appropriate because **"increases"** is the main verb of the sentence. The subject is **"growth"**, which is singular, so the verb should be **"increases" (VBZ)**.

2. Analyze the Initial Tagging Errors

The initial tagging is:

economic/JJ growth/NN increases/NNS employment/NN

The word **"economic"** is correctly tagged as **JJ** because it describes the noun **"growth."**

"growth" is correctly tagged as **NN** because it functions as the subject of the sentence.

"increases" is incorrectly tagged as **NNS**. NNS represents a plural noun, but in this sentence **"increases"** is used as a verb.

"employment" is correctly tagged as **NN** because it is the object affected by the verb **"increases."**

The grammatical structure is:

Subject → Verb → Object

growth → increases → employment

Since **"growth"** is singular, the appropriate third-person singular present-tense verb is:

increases/VBZ

Therefore, the corrected sentence is:

economic/JJ growth/NN increases/VBZ employment/NN

3. Word Frequency Distribution

Given frequencies:

- economic = 120
- growth = 450
- increases = 210
- employment = 380

First, calculate the total frequency:

Total = 120 + 450 + 210 + 380

Total = 1160

Now calculate the probability of each word:

P(economic) = 120 / 1160

P(economic) ≈ 0.1034

P(growth) = 450 / 1160

P(growth) ≈ 0.3879

P(increases) = 210 / 1160

P(increases) ≈ 0.1810

P(employment) = 380 / 1160

P(employment) ≈ 0.3276

Therefore, the word frequency distribution is approximately:

economic → 10.34%

growth → 38.79%

increases → 18.10%

employment → 32.76%

Frequency information helps probabilistic POS tagging because words that occur frequently provide stronger statistical evidence. However, frequency alone cannot always determine the correct POS tag. **Context and grammatical relationships** are also important.

For example, **"increases"** can potentially be interpreted differently depending on context. The surrounding words help determine whether it functions as a verb or noun.

4. Transformation-Based Tagging and Entropy

Transformation-Based Tagging, such as the **Brill Tagger**, first produces an initial POS tagging and then applies learned correction rules.

Initially:

economic/JJ growth/NN increases/NNS employment/NN

After applying the transformation rule:

economic/JJ growth/NN increases/VBZ employment/NN

Thus, the incorrect **NNS** tag is corrected to **VBZ**.

Transformation-Based Tagging improves accuracy because it uses **context-dependent rules** to correct errors made during the initial tagging stage.

For example:

NN + NNS → NN + VBZ

This rule recognizes that when a singular noun is followed by a word tagged as NNS in this context, the following word may actually be a third-person singular verb.

Role of Entropy

Entropy measures the uncertainty of a probability distribution.

The formula is:

$$H = -\Sigma P(tag) \log_2 P(tag)$$

Before applying the rule, the system may have uncertainty about whether **"increases"** is:

NNS or VBZ

After applying the learned transformation rule, the system has stronger evidence that:

increases → VBZ

Therefore, the uncertainty associated with the POS assignment decreases.

High entropy → High uncertainty

Low entropy → High confidence

Thus, entropy can be used to evaluate whether the transformation rules make POS tagging more confident and reliable.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________